

9.2.9 Computer programs

For the technical character of computer programs, see also chapter [1.A.2.4](#) "Computer-implemented inventions".

Purely conceptual aspects of software design and development will normally not contribute to an inventive step according to the jurisprudence of the boards of appeal ([T 983/10](#); see e.g. [T 49/99](#), [T 1171/06](#) and [T 354/07](#)).

In [T 1173/97](#) (OJ 1999, 609) the board held that a computer program product is not excluded from patentability under [Art. 52\(2\)](#) and (3) EPC 1973 if, when it is run on a computer, it produces **a further technical effect** which goes beyond the "normal" physical interactions between program (software) and computer (hardware)" (see point 9.4 of the Reasons).

In [T 2825/19](#) the board considered that opinion [G 3/08](#) of 12 May 2010, when compared to decision [T 1173/97](#), reframed the interpretation of technicality with respect to computer programs. [G 3/08](#) (Reasons 13.5 and 13.5.1), explicitly rejected the position that any technical considerations are sufficient to confer technical character on claimed subject-matter. Such a narrower interpretation of the term "technical" with respect to computer programs is a normal development for the interpretation of a legal provision open to interpretation (see opinion [G 3/19](#)), and this is the case for programs for computers "as such" in [Art. 52\(2\)\(c\)](#) and (3) EPC. The board saw no support for the appellant's view that the concept "further technical considerations" should be interpreted with a broader meaning that would also cover considerations aiming to solve problems "merely" relating to programming such as maintainability, re-usability and understandability of program code, or, in this case, the use of a universal template for translating natural language into executable expressions in external operational environments. Such a broader view of the concept "further technical considerations" appeared problematic with regard to the imperative to ensure legal certainty and judicial predictability requiring a uniform application of the law (see opinion [G 3/08](#), Reasons 7.2.3) since no criteria were apparent which could then be used to establish a clear boundary between "technical" and "non-technical" aspects of computer programs.

In [T 1370/11](#) the board stated that the improved speed of a computer program is not by itself a technical contribution to the art (see also [T 42/10](#)). In that respect, it must be established that a computer-implemented method or a computer program has a "further" technical effect and solves a technical problem independently of its absolute or relative computing time. Only then, and only if the alleged speed-up affects an established technical effect, can it be argued that the speed-up contributes to a technical effect and thus to inventive step ([T 641/00](#)).

In [T 354/07](#) the board stated that conceptual processes and meta methods for software production generally have no technical features relevant for patentability and thus cannot provide a basis for inventive step, unless, in an individual case, a direct causal connection can be proved with a technical effect which is relevant to the solution of a technical problem. Software development and manufacture takes place in several stages from

demand analysis through various design phases to production. All these stages essentially involve intellectual activity comparable to an engineer's design work, even if it is supported by programming tools and what is being designed is a technical system. While the design and programming of complex systems in particular require the active involvement of engineers and the application of technical knowledge, the eventual result to which each of these development phases is directly geared is not the technical solution of a technical problem but a requirements specification, a data, process and/or function model, or a program code. This assessment applies in particular to meta methods, which are concerned with the process of software production itself on an even more abstract level. For instance, they give the software developer instructions as to how the design process should be structured and organised or what modelling methods are to be used.

In T.1539/09 the invention related to a graphic programming language and environment that was designed to allow a user to write code without much training or expertise. The board took the view that reducing a programmer's intellectual effort was not a technical effect. This applied all the more since the effect was likewise sought for all programs developed, regardless of their purpose (see T.741/11). The board stated that the activity of programming – in the sense of writing code – is a mental act, at least to the extent that it is not used in an actual application or environment to produce a technical effect. Therefore the definition and provision of a programming language does not per se contribute to a technical solution, even if the selection of a programming language helps to reduce the programmer's intellectual effort (see also T.2270/10).

In T.42/10 and T.1281/10 claim 1 defined a method which, based on outcomes of games, calculated indications of the skills of the players by passing messages between nodes of a factor graph. The board had to determine to what extent the features of the claim had a technical character and could contribute to inventive step. The board referred to the decision of the Court of Appeal of England and Wales in *Re Gale's Application* [1991] RPC 305. The board's approach to assessing questions of what is and what is not technical about a computer-implemented method was, in this case, to ask the same questions as Nicholls LJ in *Re Gale's Application*. The first was what does the method as a whole do, and does it produce an overall technical result? The second was; if there is no overall technical result, does the method at least have a technical effect within the computer? If both questions were answered in the negative, no technical problem had been solved and there could be no inventive step. The board's view regarding technicality could be summarised as follows; the overall aim of keeping players interested is not technical. The intermediary aim of assessing and comparing playing performance is not technical. The representation of performance by probability distributions, and the updating of them, are mathematical methods. The use of factor graphs with message-passing is a matter of mathematics or abstract computer science. The board concluded that the only technical feature defined in this claim was the (computer) processor. The subject matter of claim 1, therefore, did not involve an inventive step if it would have been obvious to the skilled person, who had the task of implementing the method, to use a computer processor.

In T.1630/11 a method for simulating a multi-processor system in an electronic device was found to lack inventive step. The appellant had argued that the invention enabled users

"to efficiently model and simulate a multi-processor system". The board noted that the major part of claim 1 was concerned with the expressions of a graphical programming environment which were found in T 1539/09 (point 5 of the Reasons) not to contribute to inventive step (see also T 2270/10, point 7 of the Reasons). The board followed its earlier jurisprudence according to which modifications to a programming language or system that enable the programmer to develop a program with greater ease and thus, presumably, speed and accuracy, do not make a technical contribution to the art.

In T 2147/16, the board held that the mere assumption that an algorithm is optimised for the computer hardware and may have a technical contribution is not sufficient. The implementation of an algorithm in a method for filtering spam messages must have a proven further technical effect or specific technical considerations; such further technical effect must be specifically and sufficiently documented in the disclosure of the invention and be reflected in the claim wording; the algorithm must serve a technical purpose.

9.2.10 Assessment of features relating to a presentation of information

When deciding if a feature relating to the presentation of information is technical or not, what has to be considered is whether or not it contributes to solving a technical problem. The fact that mental activities are involved does not on its own render the subject-matter non-technical (T 643/00, T 336/14). However, a feature that solely addresses a user's **subjective preferences** does not solve a technical problem (T 1567/05).

For further information on the technical character of features relating to the presentation of information and display of data, see also chapter I.A.2.3. "Aesthetic creations" and I.A.2.6. "Presentation of information".

a) Data visualisation

The jurisprudence of the boards of appeal provides little support for the idea that screen representations inherently address technical problems (T 1143/06, T 95/86). A feature which relates to the manner in which cognitive content is conveyed to the user on a screen normally does not contribute to a technical solution to a technical problem. An exception would be if the manner of presentation can be shown to have a **credible technical effect** (T 1143/06, T 1575/07, T 1741/08 and T 1562/11). Features aimed exclusively at **improvements regarding the way information is perceived or processed** by the human mind are regarded as non-technical (see e.g. T 1567/05, T 125/04, T 579/11).

In T 543/14 the board held that providing a visual indication of technical conditions of a machine is a technical feature, according to the case law of the boards of appeal (see T 528/07, T 781/10 and T 887/12).

In T 1567/05 the board stated that the indication of "strength levels" in the form of predetermined display colours – which is a classification – has no technical effect. Although they refer here to technical phenomena, the stress values are mere pieces of information aimed exclusively at the human mind. As was noted in decision T 154/04 (point 8 of the Reasons), this list covers subject-matters whose common feature is a

substantial lack of technical character. That this is true for presentations of information was observed in decision T 119/88 (OJ 1990, 395), which states in point 4.2 of the Reasons that the classification of objects by colour represents a non-technical effect.

In T 726/07 the feature in question was the **use of a colour** to represent a state of the cache. The board stated that even assuming that this could be considered as representing "conditions prevailing in an apparatus" in the sense of T 115/85 (OJ 1990, 30), the board considered that the **use of a colour** was a common and obvious implementation of a status indication.

In T 1734/11 the board stated that a reduction of user interaction does not necessarily convey technical character to the means for achieving the reduction (following T 1741/08). Inputting information to a machine and **reducing the burden of doing so**, while these may be technical tasks a priori (at least in the sense that they are not listed in Art. 52(2) EPC), not everything that supports a technical task has itself a technical character. However, in contrast to the presentation of the state of a technical machine, which may have technical character (T 115/85, OJ 1990, 30), the board considered that the presentation of pricing information, for example by **colour coding**, is a non-technical aspect even if it helps the user to conduct a price-sensitive travel query more efficiently.

b) Graphical user interfaces

A **GUI layout** (graphical user interface) as such is considered to be non-technical, being a "presentation of information" (Art. 52(2)(d) EPC; T 1741/08 citing T 1143/06). See also T 1214/09, T 1185/13.

Features concerning the graphic design of user interfaces do not have a technical effect, because their design is not based on technical considerations, but on general intellectual considerations as to which design is particularly appealing to a user (Guidelines G-II, 3.7.1 – March 2022 version). For example, the colour (see T 1567/05, T 726/07, T 1734/11), shape (T 677/09), size (T 823/07, T 1237/07), layout (T 756/06, T 1741/08, T 1214/09), arrangement of items on the screen (T 643/00, T 1237/10) or the information content of a message displayed is usually not a technical aspect of a graphical user interface. An exception would be if these features contribute to achieving a particular technical effect (T 1741/08, T 1143/06).

A feature defining a presentation of information produces a technical effect if it credibly assists the user in performing a technical task by means of a continued and/or guided human-machine interaction process (T 336/14, T 1802/13 and T 1185/13). Such a technical effect is considered credibly achieved if the assistance to the user in performing the technical task is objectively, reliably and causally linked to the feature. This would not be the case if the alleged effect depends on subjective interests or preferences of the user (Guidelines G-II, 3.7 – March 2022 version).

In T 1235/07 the board took a wider view of "presentation of information" than just the actual information that is displayed, the so-called cognitive content, to include also structural aspects of how the information is displayed. This broader meaning was

supported by almost the only reference to the presentation of information in the travaux préparatoires. In the board's view, such additional aspects can only contribute to inventive step if they have technical character. The board considered the circumstances of the case in hand to be similar to those in T 1143/06, but they were even less convincing because the invention was merely a visual display of a known analysis technique using known means, whereas in T 1143/06 the idea of varying the speed of the element was at least not known.

i) Broken technical chain

In T 1741/08 the application was concerned with a method of entering data into a data processing system. The question was whether an additional technical effect could be attributed to a particular layout of a graphical user interface (GUI). The board made clear that "lowering the cognitive burden of the user" is not of itself a technical effect (T 1143/06 followed, T 49/04 not followed). It found that the reduction in use of resources would be caused by the way the brain of the user perceives and processes the visual information given by a particular way of presenting information. The appellant also argued that there was a chain of effects. But in terms of technical effects this was a broken chain. The board did not accept that such a broken chain could be used as evidence of the required technical effect overall ("broken-technical-chain fallacy"; see also T 158/88, T 603/89 and T 1670/07). Following T 1143/06, the board considered a **GUI layout as such to be non-technical**, being a "presentation of information" (Art. 52(2)(d) EPC). In the case at issue, the **arrangement of icons** displayed was intended to convey information, namely at which step in the input process the user found him- or herself. The board stated that the mere fact that a particular **choice of information to display or of how to display** it is particularly clear, lucid, or "lowers the cognitive burden" of the user was not sufficient to demonstrate that the choice has a technical effect. The case law of the boards of appeal was consistent with this principle. The fact that not all GUI-related applications are considered to be inventive does not mean that there is a contradiction between the decisions.

In T 1670/07 the board confirmed the "broken-technical-chain fallacy" (T 1741/08) and cited T 603/89 and T 1670/07 as examples. The board held that the possible final technical effect brought about by the action of a user could not be used to establish an overall technical effect because it was conditional on the mental activities of the user.

In T 1214/09 turned on whether **improving a human's cognitive evaluation** of certain information by changing the manner in which the information is presented was non-technical. The board noted that the alleged effect of feature (i), i.e. the improved evaluation of thumbnail file images by the user, was due solely to the claimed arrangement of thumbnail file images. In the board's judgment, this arrangement was not based on considerations other than those proper to the field of designing presentations of information for human viewing and was hence not an expression of any technical principle. The board therefore considered the alleged effect not to be a technical effect since in the context of this invention any **improvement** in the efficiency of image retrieval can only be the result of the non-technical improvement in the user's evaluation of the displayed thumbnail file images.